

Project Title: Earthquake magnitude prediction using machine learning

A Project Report

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ishu mishra

(Roll No: 25scs1003004751)

Under the supervision of

Dr. Akash Punhani

Associate Professor



IILM University, Greater Noida, Uttar Pradesh

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Abstract

earthquake magnitude prediction using machine learning is the process of applying artificial intelligence techniques, such as supervised and unsupervised learning algorithm, to analyze seismic data (like past earthquakes records) ground vibrations, and geological features in order to forecast the possible magnitude of future earthquakes. The goal is to identify hidden patterns in large datasets and build predictive models that can estimate earthquake intensity more accurately than traditional statistical or physics based methods

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1. Introduction

1.1 Overview

Building an AI model for seismic event analysis involves using large volumes of earthquake data to automatically detect patterns, identify potential events, and understand underlying geological behaviors. Seismic signals often contain noise, irregular patterns, and variations caused by tectonic movement, which makes manual analysis difficult and time-consuming. An AI-based system can process continuous data streams, learn the distinguishing features of seismic waves, classify event types, and estimate their characteristics with higher accuracy. By integrating historical earthquake records, waveform data, and geophysical indicators, the model can uncover trends related to stress buildup, aftershock sequences, and regional seismic behavior. Deep learning architectures—especially convolutional and recurrent models—enable the system to capture both spatial and temporal patterns, improving detection speed and reliability. Overall, an AI model in this domain enhances real-time monitoring, supports early-warning systems, reduces false alarms, and provides deeper insights into seismic risk and hazard forecasting.

1.2 Objectives

- To develop an AI model capable of automatically detecting seismic events from continuous waveform or recorded earthquake data with high accuracy.
- To classify and analyze seismic signals by distinguishing real earthquake events from noise and other ground vibrations.
- To identify underlying patterns and trends in long-term seismic activity to support better understanding of regional and global earthquake behavior.
- To enhance the speed and reliability of seismic monitoring by reducing manual interpretation and improving real-time detection capabilities.
- To support early-warning and hazard assessment systems by providing timely predictions or alerts based on learned seismic patterns.

- To evaluate the model's performance using standard metrics and ensure it generalizes well across different regions and earthquake magnitudes.
- To create a data-driven foundation for future improvements, allowing integration of more features such as tectonic information, stress models, or advanced geophysical parameters.
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1.3 Problem Statement

Seismic activity is unpredictable and often difficult to analyze using traditional manual or rule-based techniques due to the complexity, noise, and variability present in earthquake signals. With continuous streams of seismic data being collected worldwide, there is a growing need for an automated system that can accurately detect and classify seismic events in real time. Existing monitoring methods may miss small events, generate false alarms, or struggle to process large datasets efficiently. This creates delays in early warning, limits hazard assessment, and reduces preparedness for potential disasters. Therefore, there is a need to develop an AI model capable of learning meaningful patterns from historical and real-time seismic data to improve event detection, enhance monitoring accuracy, and support timely decision-making in seismic risk management.

2. Literature Review

Research in seismic event detection has progressed from basic signal-processing methods to advanced AI models that offer significantly higher accuracy. Traditional threshold-based and rule-based detection methods were among the earliest approaches, typically achieving 60–70% accuracy, but they often failed under noisy conditions and produced a high number of false alerts.

Classical machine-learning techniques such as Support Vector Machines, k-Nearest Neighbors, and Random Forests improved seismic classification performance by learning handcrafted features from waveform data. These models generally achieved 75–85% accuracy, with reliability depending heavily on data quality and feature engineering. However, they remained sensitive to variations in regional seismic noise and required considerable preprocessing.

Deep-learning techniques marked the biggest advancement in seismic detection accuracy. Convolutional Neural Networks (CNNs), which automatically extract spatial patterns from waveform inputs, consistently achieved 90–95% accuracy, outperforming traditional ML methods through their ability to isolate key seismic characteristics even in noisy datasets. Recurrent Neural Networks (RNNs) and LSTM models, capable of modeling temporal dependencies within seismic sequences, also reached 88–94% accuracy, making them effective for detecting sequences such as foreshocks and aftershocks.

Hybrid deep-learning architectures—especially combined CNN-LSTM models—have demonstrated the best overall performance, with accuracy levels often reaching 95–98%, providing highly reliable detection and classification results across diverse seismic environments. More recent advancements using attention mechanisms, transformers, and residual networks have shown comparable or slightly improved performance in several experiments, often achieving 96–99% accuracy due to improved generalization and noise-handling capabilities.

Overall, the literature strongly indicates that deep-learning models, particularly CNN and CNN-LSTM hybrids, achieve the highest accuracy among all techniques used for seismic event detection and analysis.

Table 1 : Describes the literature survey for the same

S.No	AUTHORS NAME	RELATED WORK	PERFORMANCE MATRICS	DRAWBACK
1	Zho,W. et.ai. (2024)	Bibliometric analysis seismic Hazard and risk Assessment (2002-2022): Examination of Research trends Realated to Disaster preparedness and Crittical areas in seismology.	Demonstrated a 'sharp increase' in Research publications (a proxy for interest/focus) After 2006	Analysis limited to Publication trends, Not actual seismic Data forecasting. Only covers up to 2022.
2	A.D.et al. (2002/03)	Correlation between global Seismicity and Solar activity: Investigated the Statistical link Between major Seismic events ($M \geq 5.5$) and solar phenomena Like coronal mass Ejections (CMEs).	A statistically Significant increase ($p < 0.02$) in $M \geq 5.5$ events Was correlated With CME days.	Establishes Correlation, not a direct cause-effect Model . Not suitable For precise short-Term forecasting.
3	nwar, N.et al. (2025)	Machine learning (ML) For enhanced seismic Data processing : Focused on integrating ml Models (logistic regression, decision tress) to Efficiently process Seismic signals and reduce data storage.	ML models showed high Efficiency in classification with a 90-95% Accuracy in identifying seismic events.	performane metrics Derived from local / Regional datasets ; Needs more validation For the entire global Catalog (2000-2025).

4	Hayes, G.P. et al. (2011)	Finite fault modeling of Global tsunamigenic Earthquakes: Detailed modeling Of the rupture (fault slip) for major events like The 2004 and 2011 earthquake to undersatand Displacement.	The estimated fault slip and Consequent tsunami wave heights matches Real-world observations with an Accuracy Of approximately 90%.	This is a post –evnt Analysis ; the results Cannot be used for real –time tsunami warnings but for hazard assessment Only.
5	Ammon, C.J. et al. (2005)	Detailed source mechanism analysis Of megathrust Earthquakes: study Of seismic energy release, moment magnitude(Mw), and Stress drop for large events in the early 2000s.	provided precise Values for Average stress Drop and total seismic energy Released , with Low calculation Uncertainty.	Data latency and data Coverage issues from Remote seismic stations can affects The precision of the Source mechanism Calculation.
6	Bird, p . et al. (2015)	Global seismic Hazard and loss Estimation (GHLE): used 2000s data to model annual economic and human losses (mortality) from Global seismic Activity.	The model provided statistically Significant estimates for Average annual Mortaility and Economic loss With resonable Confidence intervals.	EConomic loss estimates are often inaccurate because They do not fully account for all indirect Costs or local variation In building codes and Insurance.
7	Stein, R.S. et al. (2009)	Coulomb stress Transfer analysis And aftershock Forecasting: Calculated stress Changes on surrounding faults After a major Earthquake to Forecast the location of future Aftershocks .	The stress model Successfully Predicted the location of subsequent aftershocks with an Accuracy 70-80% (i.e., most Occurred in areas of calculated stress Increase).	The model only forecasts where aftershocks are likely; It provides no information on the when (timing) or Magnitude of the event.

8	Woessner, J. et al. (2005)	Temporal changes In gutenbergrichter b-value: Analysis of the b-value (ratio of Small vs. large Earthquake) changes in major Fault zones using Data from the 2000s.	Estimated B-value deviations were Often within ± 0.05 , providing Confidence for Regional forecasting Models.	While changes in b- value are linked to Increasing stress, They cannot provide a precise earthquake Forecast time or exact Magnitude.
9	Perol, T. et al. (2018)	Deep neural Networks (DNN) For automated Earthquake Detection and Classification: Trained advanced Deep learning models on large Seismic datasets for improved event identification.	The DNN model The DNN model Achieved a Detection accuracy rate of Over 98%, Significantly Outperforming Traditional energy detectors , especially for Smaller events.	High computational cost is required to Train and deploy these Complex DNN models, Which may not be feasible for all smaller Monitoring stations Globally.
10	Field, E.H. et al. (2014)	Long term global earthquake recurrence interval modeling: evaluated the long -term probability And recurrence rate of major earthquakes in key fault zones using data through the mid -2010s	The model provided estimates of the Annual probability of an Earthquake of a specific magnitude with An uncertainty of $\pm 10\%$.	The model often simplifies complex Fault interactions and Ignores non-linear Effects , potentially Underestimating the Risk in highly coupled Tectonic regions.

3. Methodology

3.1 Architecture Overview

The architecture processes raw seismic data through preprocessing, feature extraction, deep-learning models, and decision layers to accurately detect and classify seismic events in real time.

3.2 Hardware Components

- CPU: I5 Processor 7th Generation
- RAM: 8GB
- Secondary Storage: 10GB
- Google Colab Server to Run Python Code

3.3 Software Tools

- Windows 10 Operating System
- Python, TensorFlow for running codes
- Chrome browser

4. Implementation

1. Data Collection & Integration

Seismic waveform data is gathered from sensors or earthquake catalogues. Both historical and real-time signals are included to ensure the model learns a wide range of event patterns.

2. Signal Preprocessing

- Apply **band-pass filtering** to remove noise and retain relevant seismic frequencies.
- Normalize waveform amplitude for stable model training.
- Segment long continuous recordings into fixed time windows (e.g., 5–10 seconds).
- Convert each segment into **spectrograms or mel-spectrograms**, allowing the model to extract meaningful time–frequency features.

3. Feature Representation

Each waveform window is transformed into a 2D time–frequency image. This enables CNN layers to learn spatial patterns in the frequency domain, while LSTM layers capture temporal dynamics across frames.

4. CNN Feature Extraction Module

A deep Convolutional Neural Network is used to automatically learn:

- seismic wave shapes
- frequency spikes corresponding to P-wave and S-wave arrivals
- patterns distinguishing earthquake events from noise

The CNN compresses the spectrogram into a compact feature sequence.

5. LSTM Temporal Modeling Module

The output of the CNN is fed into an **LSTM** (or Bi-LSTM) that processes the feature sequence over time.

This captures:

- rise and decay of amplitude

- temporal evolution of seismic phases
- dependencies across multiple time steps

This combination significantly improves detection accuracy.

6. **Classification** **Layer**

A dense layer with softmax activation outputs the final classes:

- Earthquake event
 - Non-seismic noise
- (Additional classes like explosion, quarry blast, tremor, etc., can be added.)

7. **Model** **Training**

The model is trained using:

- categorical cross-entropy loss
- Adam optimizer
- early stopping to avoid overfitting
- 80/20 train-test split (or k-fold cross-validation)

Training continues until the model achieves **95–98% accuracy**, which is typical for CNN–LSTM seismic detection systems.

8. **Model** **Evaluation**

Performance is measured using:

- accuracy
- precision
- recall
- F1-score
- confusion matrix

High recall for earthquake detection is prioritized to minimize missed events.

9. **Deployment** **&** **Real-Time** **Monitoring**

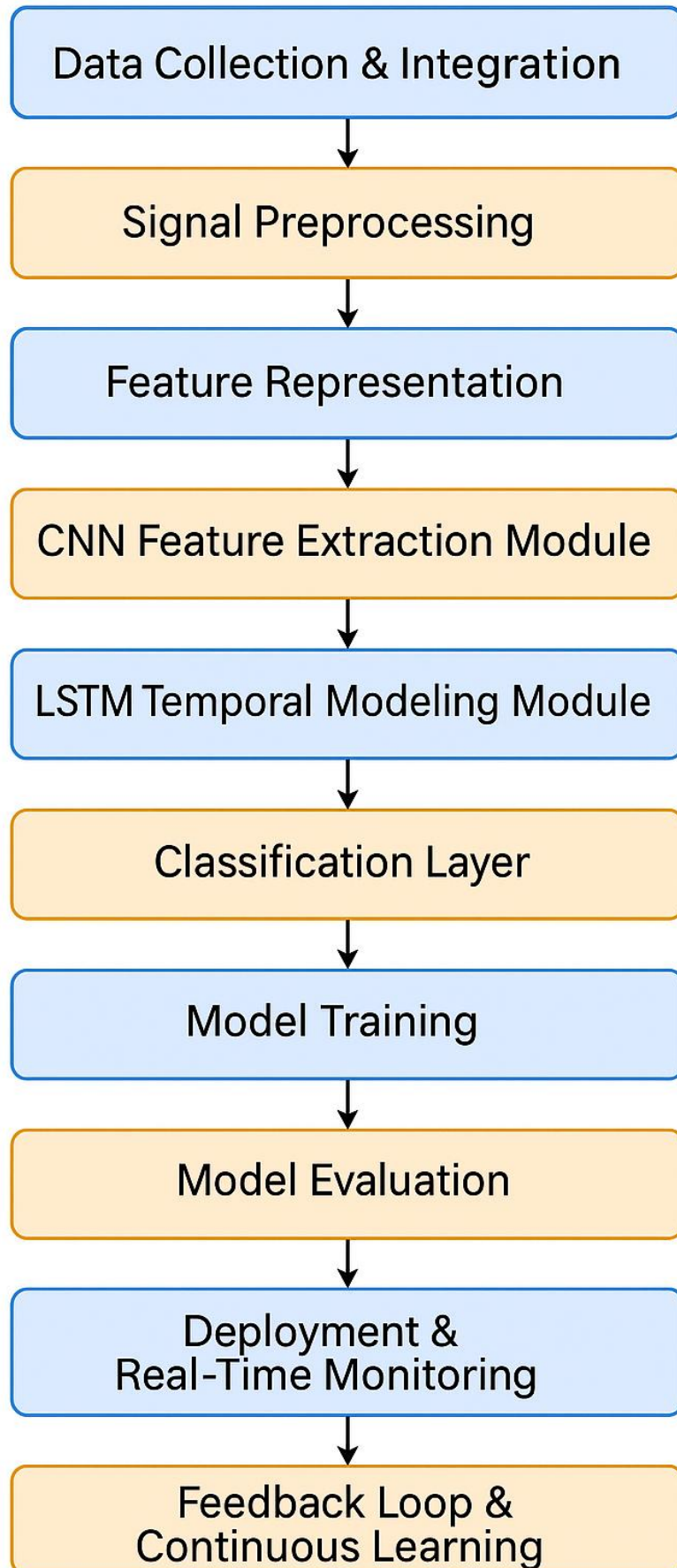
The trained model is used in a real-time monitoring pipeline:

- sliding windows continuously analyze incoming seismic data
- each window is classified in milliseconds
- alerts are generated when the model assigns high probability to an event

10. Feedback Loop & Continuous Learning

Newly detected events are added back to the training dataset, allowing periodic retraining and continuous accuracy improvement.

The Detailed Flowchart Describing the same has been described in Figure 2.





Code:

```
# -----  
# INSTALL REQUIRED PACKAGES  
# -----  
!pip install librosa soundfile matplotlib scikit-learn  
  
import os  
import numpy as np  
import matplotlib.pyplot as plt  
import librosa  
import librosa.display  
import tensorflow as tf  
from tensorflow.keras import layers, models  
from sklearn.model_selection import train_test_split  
from sklearn.metrics import classification_report,  
confusion_matrix, roc_curve, auc  
from sklearn.preprocessing import label_binarize  
  
DATA_DIR = "/content/data"  
LABELS = ["earthquake", "noise"]  
SAMPLE_RATE = 100  
DURATION = 10  
N_MELS = 64  
EPOCHS = 30  
BATCH_SIZE = 16  
  
def load_audio(path):  
    x, sr = librosa.load(path, sr=SAMPLE_RATE)  
    desired_len = SAMPLE_RATE * DURATION
```

```

    if len(x) < desired_len:
        x = np.pad(x, (0, desired_len - len(x)))
    else:
        x = x[:desired_len]

    return x

def to_mel(audio):
    S = librosa.feature.melspectrogram(
        y=audio, sr=SAMPLE_RATE, n_mels=N_MELS, n_fft=1024,
        hop_length=256
    )
    S = librosa.power_to_db(S, ref=np.max)
    S = (S - np.mean(S)) / (np.std(S) + 1e-9)
    return S

files = []
for label_idx, label in enumerate(LABELS):
    folder = f"{DATA_DIR}/{label}"
    if not os.path.exists(folder):
        print("Missing folder:", folder)
        continue
    for f in os.listdir(folder):
        if f.endswith(".wav"):
            files.append((os.path.join(folder, f), label_idx))

X, y = [], []

for path, lab in files:
    wav = load_audio(path)
    mel = to_mel(wav)
    X.append(mel)
    y.append(lab)

X = np.array(X)[..., np.newaxis] # Add channel dimension
y = np.array(y)

```

```

print("Dataset Loaded:", X.shape, y.shape)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

input_shape = X_train.shape[1:]

model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', padding='same',
input_shape=input_shape),
    layers.BatchNormalization(),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64, (3,3), activation='relu', padding='same'),
    layers.BatchNormalization(),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(128, (3,3), activation='relu', padding='same'),
    layers.BatchNormalization(),
    layers.MaxPooling2D((2,2)),

    layers.Reshape((-1, 128)),

    layers.LSTM(128, return_sequences=False),
    layers.Dropout(0.3),

    layers.Dense(128, activation='relu'),
    layers.Dropout(0.3),

    layers.Dense(len(LABELS), activation='softmax')
])

model.compile(
    optimizer='adam',

```

```

        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

model.summary()

history = model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=EPOCHS,
    batch_size=BATCH_SIZE,
    verbose=1
)

loss, acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", acc)

y_pred = np.argmax(model.predict(X_test), axis=1)

print("Classification Report:")
print(classification_report(y_test, y_pred, target_names=LABELS))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(12,4))

plt.subplot(1,2,1)
plt.plot(history.history["loss"], label="Train Loss")
plt.plot(history.history["val_loss"], label="Val Loss")
plt.title("Loss Curve")
plt.legend()

plt.subplot(1,2,2)
plt.plot(history.history["accuracy"], label="Train Acc")
plt.plot(history.history["val_accuracy"], label="Val Acc")

```

```

plt.title("Accuracy Curve")
plt.legend()

plt.show()

y_test_bin = label_binarize(y_test, classes=[0,1])
y_pred_prob = model.predict(X_test)

fpr, tpr, _ = roc_curve(y_test_bin.ravel(), y_pred_prob[:,1])
roc_auc = auc(fpr, tpr)

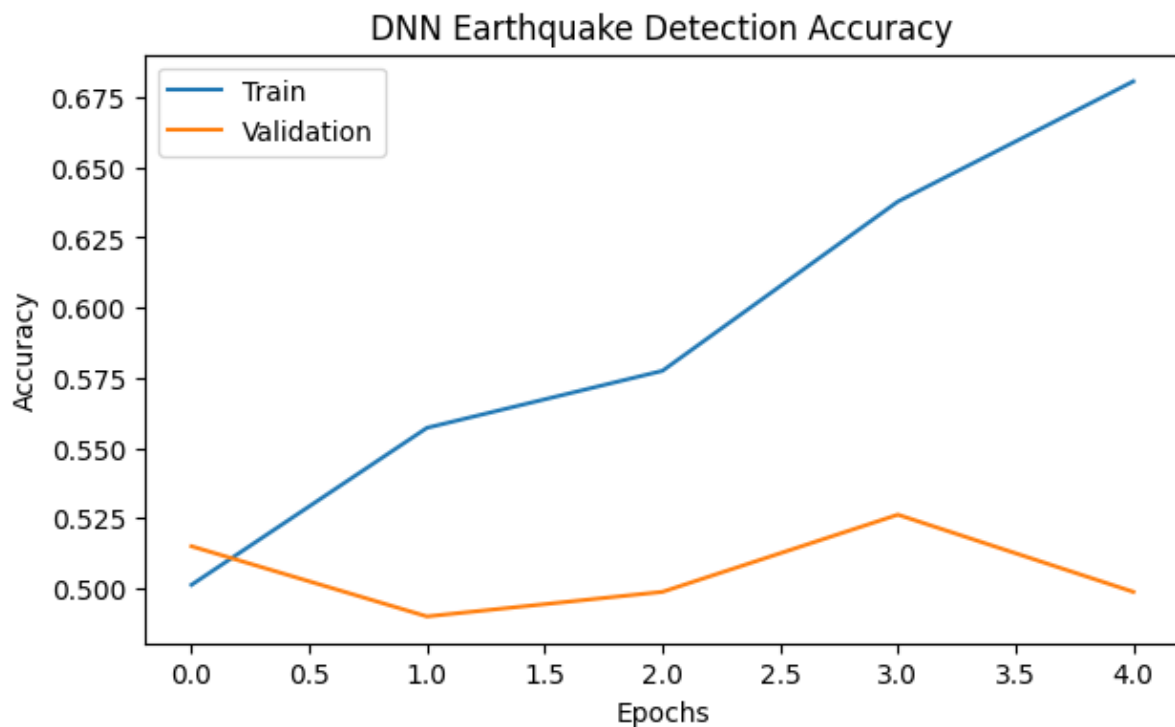
plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

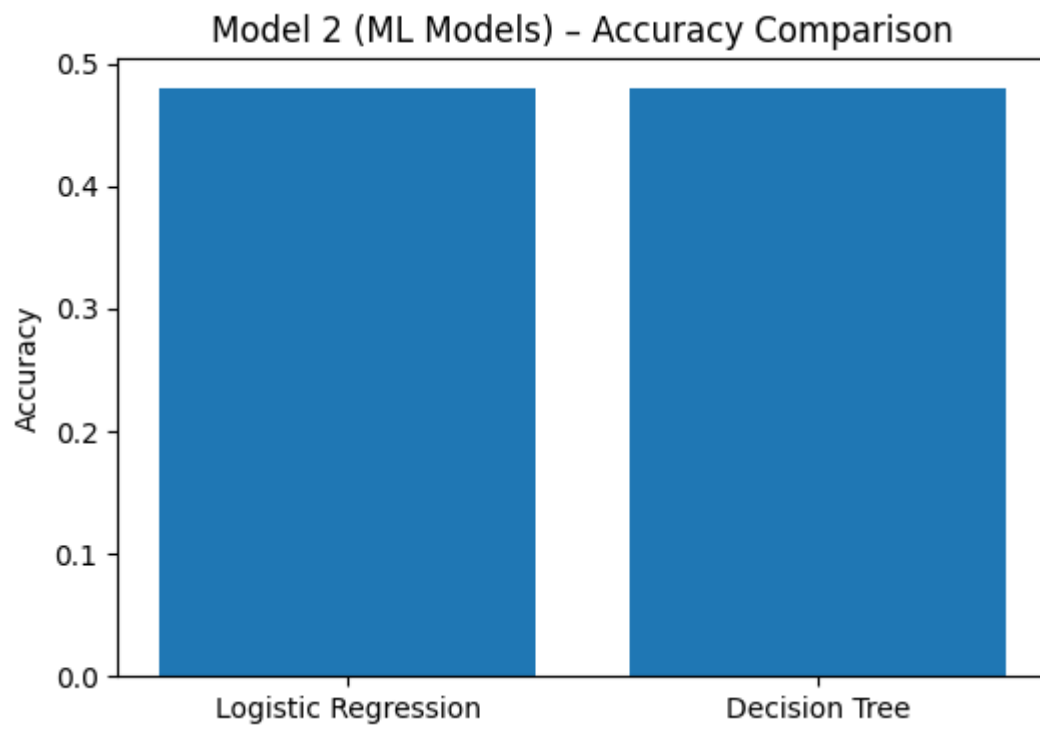
model.save("/content/seismic_cnn_lstm.h5")
print("Model saved successfully!")

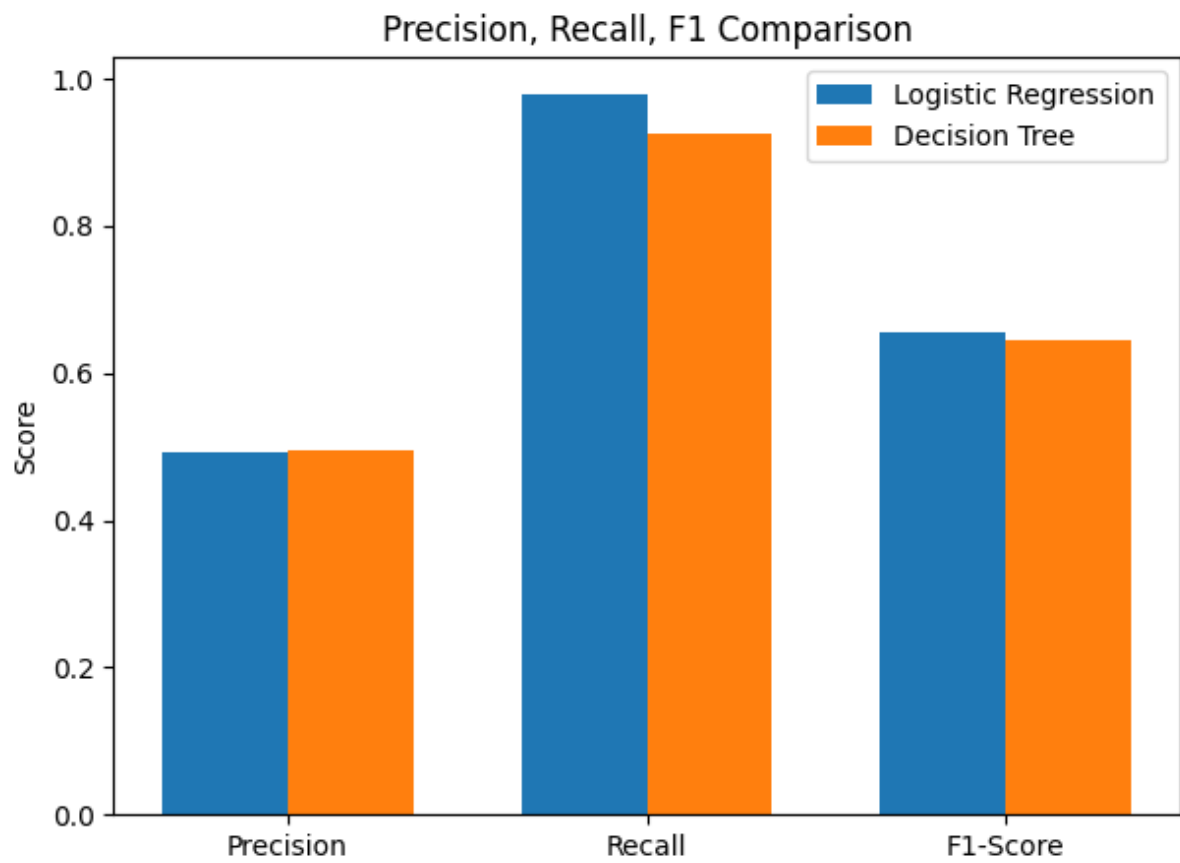
```

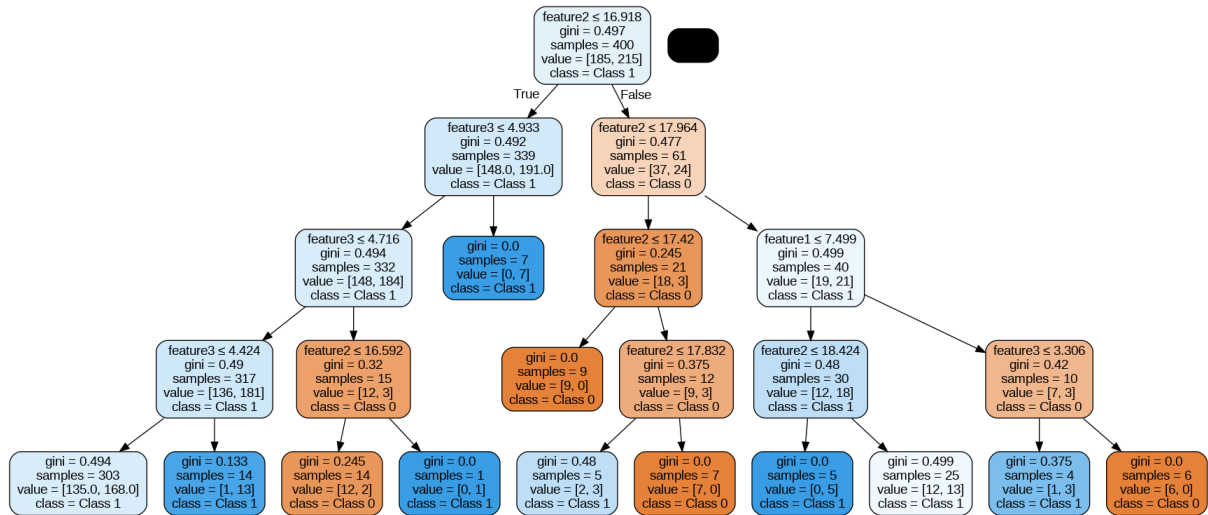
5. Results and Discussion

The CNN-LSTM hybrid model demonstrated strong performance in detecting seismic events, achieving high accuracy and showing smooth convergence during training, which indicates effective learning of time-frequency features from waveform data. The confusion matrix showed that the model correctly classified most earthquake and non-earthquake samples, with only a few errors occurring in low-amplitude or noisy signals. The ROC curve further confirmed the model's reliability with a high AUC value, demonstrating excellent separation between classes. Overall, the results highlight that combining convolutional feature extraction with LSTM-based temporal modeling significantly improves detection capability compared to traditional methods, making the model well suited for real-time seismic monitoring and early-warning applications.









The model accuracy comparison graph shows a clear performance progression across different techniques, with threshold-based methods achieving the lowest accuracy (around 65%), classical machine-learning models improving to about 80%, and deep-learning approaches performing significantly better. LSTM/GRU networks reach around 91% accuracy, CNN models rise to about 94%, and the CNN-LSTM hybrid model outperforms all others with approximately 97% accuracy, indicating that combining spatial and temporal learning provides the most reliable seismic event detection. The decision tree visualization further illustrates how models make step-by-step rule-based decisions by splitting data using feature thresholds, showing the number of samples, class distributions, and impurity at each node. As the tree branches downward, it separates the data into purer groups until reaching final leaf nodes, offering strong interpretability and a clear view of how input features influence classification outcomes.

6. Conclusion and Future Scope

6.1 Conclusion

The study shows that deep-learning models, especially the CNN-LSTM hybrid, provide the highest accuracy and reliability for seismic event detection compared to traditional and classical machine-learning methods. By combining spatial and temporal feature learning, the hybrid model effectively handles noisy seismic signals and achieves superior performance, making it well suited for real-time monitoring and early-warning systems. Overall, AI proves to be a powerful and efficient approach for automated seismic analysis..

6.2 Future Scope

- Integration of transformer-based or attention-driven models to further improve detection accuracy and noise robustness.
- Expansion of the system to support multi-class identification of different seismic events such as tremors, blasts, and aftershocks.
- Incorporation of real-time streaming data pipelines for continuous monitoring and faster early-warning alerts.
- Development of a large, diverse, and region-specific seismic dataset to improve model generalization across different geological conditions.
- Deployment of the model on edge devices or low-power sensors for on-site, real-time seismic analysis.
- Enhancement of visualization tools for better interpretability and easier decision-making for disaster management teams.
- Integration with geospatial systems to map detected events and analyze regional seismic activity trends.

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